

Big Bang Nucleosynthesis and the Primordial Lithium Problem

Hydrogen, helium and lithium production from standard-network controls to a finite-temperature selective-transfer candidate

Standalone technical paper and research status report

Clive Stewart Boyd

21 August 2026 | Melbourne, Australia

EVIDENCE STATUS		
Current	claim	boundary
Standard BBN remains successful for deuterium and helium at the adopted baryon density.		
The primordial lithium discrepancy remains unresolved: standard-network calculations predict roughly three to four times the warm metal-poor stellar plateau.		
July 2026 expansion, clock-proxy, reaction-rate and SU(2)-thermodynamic controls produced no joint deuterium-helium-lithium solution.		
The August programme identified a narrow post-BBN electromagnetic-transfer candidate and qualified its cascade, production, FIRAS and leading finite-temperature source calculations.		
Scientific promotion remains false. The full Planck PR4 nuisance likelihood is still running; matched next-to-leading-order thermal corrections and independent reproduction also remain mandatory.		

Abstract

Big Bang nucleosynthesis (BBN) predicts the primordial abundances of deuterium, helium-3, helium-4 and lithium-7 from a compact set of early-Universe and nuclear inputs.

Observed deuterium and helium remain broadly concordant with the baryon density inferred from the cosmic microwave background (CMB), whereas metal-poor stellar atmospheres show substantially less lithium than standard BBN predicts.

This mismatch is the cosmological lithium problem.

This paper records a two-stage investigation.

The July 2026 programme used LINX to test expansion changes, effective clock shifts, neutron-lifetime controls, selected nuclear-rate pulls, network truncation, numerical convergence, a hierarchical stellar-depletion model and a published high-temperature SU(2)-CMB thermodynamic bridge.

None reduced lithium into the registered interval while preserving deuterium and helium.

The August 2026 programme therefore moved from global changes to selective post-BBN transfer.

A threshold screen showed that about 64.5% of the mass-7 reservoir must be removed to enter the adopted two-standard-deviation lithium interval.

ACROPOLIS then propagated non-thermal photon cascades through a frozen 4,800-point LINX abundance design.

A candidate near mediator mass 4.44 MeV and lifetime 1.5849×10^5 s emerged.

A production calculation based on an axion-like particle (ALP) coupled to photons, a full-spectrum COBE/FIRAS test, an independent momentum-resolved Boltzmann implementation and a leading resummed finite-temperature photon calculation retained a numerical candidate.

The present best qualified production estimate is $n_{\text{phi}}/n_{\text{gamma}} = 1.01935 \times 10^{-6}$.

Under the frozen abundance and compressed CMB likelihood it gives $D/H = 2.49454 \times 10^{-5}$, $Y_p = 0.246733$, $\text{lithium-7}/H = 1.78974 \times 10^{-10}$ and a joint-pass probability of 0.6829.

Those figures are not a claimed solution: the full Planck PR4/NPIPE nuisance likelihood, matched thermal corrections, model-wide constraints and independent reproduction remain open gates.

Headline findings

1. **Standard-background control:** Thirty thousand five hundred and three completed LINX states and a 92-evaluation validation matrix produced zero joint deuterium-helium-lithium passes.
2. **Anomaly scale:** the standard D+He+CMB predictive lithium abundance is about 5.49×10^{-10} , versus an adopted plateau likelihood centred on 1.45×10^{-10} .
3. **Mechanism selectivity:** neutron injection is excluded by the deuterium penalty, but photons between the beryllium-7 and deuterium photodisintegration thresholds can preferentially reduce mass 7.
4. **Numerical candidate:** the resummed production chain predicts $\text{lithium-7}/H = 1.78974 \times 10^{-10}$ while retaining D/H, Y_p , helium-3/deuterium, entropy and FIRAS gates in the frozen likelihood.
5. **Promotion boundary:** this is a qualified candidate, not a discovery or a solved lithium problem. Full Planck nuisance convergence, matched next-to-leading-order plasma physics and external reproduction are required.

Repository record: cliveboyd/Cosmology; historical BBN results: 2026-07-17/bbn_lithium; analysis programmes: code/bbn_lithium

Contents

- 1** Scope, terminology and evidence levels
- 2** Standard BBN foundations
- 3** Hydrogen and deuterium nucleogenesis
- 4** Helium-3 and helium-4 nucleogenesis
- 5** Lithium-7 and the mass-7 pathway
- 6** The cosmological lithium anomaly
- 7** July 2026 analysis programme
- 8** August selective-transfer gate
- 9** ACROPOLIS, LINX and the joint likelihood
- 10** Freeze-in production and spectral distortions
- 11** Finite-temperature resummation
- 12** Full Planck nuisance likelihood
- 13** Programmes, datasets and reproducibility
- 14** Notable findings and interpretation
- 15** Ongoing investigations and promotion gates
- Appendix A** Symbols and variables
- Appendix B** Acronyms
- Appendix C** Programme and data manifest
- References** Primary scientific and software sources

Navigation note

The page numbers are updated automatically by Word or LibreOffice on opening.

Equations, figures and tables are cross-referenced by their fixed report numbers.

1. Scope, Terminology and Evidence Levels

Nucleogenesis is used here in the broad historical sense of creating nuclides; nucleosynthesis is the more precise term for the reaction network that assembles light nuclei.

Hydrogen in this report includes the surviving proton reservoir and the deuterium and tritium isotopes.

Helium includes helium-3 and helium-4. Lithium refers primarily to lithium-7, most of which is produced indirectly as beryllium-7 and converted later by electron capture.

Three logically different questions are kept separate:

First, does standard BBN reproduce the observed light-element pattern at a CMB-compatible baryon density?

Second, can a parameterised mechanism move lithium without damaging deuterium or helium?

Third, does a complete microscopic model predict the necessary abundance, lifetime, transfer spectrum and CMB response without refitting them independently?

The current programme has progressed through the second question and part of the third.

Table 1. Evidence ladder used throughout this paper

Level	Meaning	What has been achieved	What is not established
E0	Analytic or dimensional screen	Threshold and entropy constraints	A physical abundance or transfer history
E1	Reproducible numerical result	LINX, ACROPOLIS, Boltzmann and FIRAS calculations	External replication
E2	Preregistered candidate mechanism	Frozen 4.44 MeV, 1.5849×10^5 s cell	Full experimental likelihood promotion
E3	Likelihood-qualified model	Compressed abundance/CMB and FIRAS gates only	Converged full Planck nuisance result
E4	Scientific claim	Not reached	Independent code/data reproduction and model-wide constraints

Interpretive rule

A lower chi-squared, a passing numerical control or a plausible particle cell is not by itself a lithium solution.

Promotion requires the same physical model to pass deuterium, helium, lithium, entropy, CMB anisotropy, CMB spectral-distortion and external-reproduction gates.

2. Standard Big Bang Nucleosynthesis Foundations

BBN operates after weak freeze-out and before the plasma becomes too cool and dilute for charged-particle reactions.

At temperatures near the mega-electronvolt scale, weak interactions set the neutron-to-proton ratio.

Nuclear assembly waits for the deuterium bottleneck to open near 0.08 MeV; most surviving neutrons then enter helium-4, while trace flows produce deuterium, helium-3, tritium, beryllium-7 and lithium-7.

The relevant epoch spans roughly seconds to tens of minutes, although late electromagnetic injection can modify the frozen abundances much later.

Equations 1-3. Baryon density and radiation-era expansion

$$\eta = n_b / n_\gamma \quad (1)$$

$$\eta_{10} = 10^{10} \eta \quad (2)$$

$$H^2 = (8\pi G/3) \rho_{\text{tot}} \quad (3)$$

Here n_b and n_γ are the baryon and photon number densities, H is the Hubble expansion rate, G is Newton's constant and ρ_{tot} is the total energy density.

Equations 4-5. Radiation approximation and CMB conversion

$$H(T) = 1.66 g^{*}(1/2) T^2 / M_{\text{Pl}} \quad (4)$$

$$\eta_{10} = 273.78 \omega_b \quad (5)$$

The conversion in Equation (5) uses $T_{\text{CMB}} = 2.7255 \text{ K}$; $\omega_b = \Omega_b h^2$ is the physical baryon density. The small conversion uncertainty is negligible for the present gate.

The reaction network evolves abundance per baryon $Y_i = n_i/n_b$.

Every reaction contributes a gain or loss term whose rate depends on temperature, density and the reactant abundances.

Nuclear-rate covariance matters because deuterium is now observed more precisely than several theoretical rate combinations are known.

Equations 6-7. Abundance evolution and Gaussian diagnostic

$$dY_i/dt = \sum_r N_{ir} \lambda_r(T) \prod_j Y_j^{v(jr)} \quad (6)$$

$$\chi^2_X = [X_{\text{pred}} - X_{\text{obs}}]^2 / (\sigma_{\text{obs}}^2 + \sigma_{\text{th}}^2) \quad (7)$$

N_{ir} is the net number of nuclides i created by reaction r , λ_r is its thermal rate and $v(jr)$ is the reactant multiplicity. Equation (7) is a transparent scalar diagnostic; the final inference uses joint covariance or posterior sampling where available.

Table 2. Adopted observational and project anchors

Quantity	Anchor used in August likelihood	Role
D/H	$(2.533 \pm 0.024) \times 10^{-5}$	Baryometer; strongest penalty against indiscriminate late injection
Y_p	0.2458 ± 0.0013	Weak freeze-out, neutron lifetime and expansion constraint
Li-7/H	$(1.45 \pm 0.25) \times 10^{-10}$	Warm metal-poor plateau likelihood used for the project gate
He-3/D	< 1.13 at the registered two-sigma gate	Checks excessive deuterium destruction and helium-3 production
N_{eff}	Sampled or compressed CMB constraint	Radiation density and photon-heating control

These anchors are analysis choices, not universal constants.

The report therefore quotes the registered likelihood inputs and separates them from newer or alternative measurements.

For context, Cooke, Pettini and Steidel reported $D/H = (2.527 \pm 0.030) \times 10^{-5}$, and the 2026 LBT helium programme reported $Y_p = 0.2458 \pm 0.0013$.

Both are consistent with standard BBN at the CMB baryon density within their stated error budgets [3,4].

3. Hydrogen and Deuterium Nucleogenesis

Most baryons remain as protons, so ordinary hydrogen is the dominant baryonic species after BBN.

The diagnostically powerful hydrogen isotope is deuterium.

At high temperature, energetic photons dissociate deuterons faster than they can survive; once the photon tail weakens, the deuterium bottleneck opens and deuterium becomes the gateway to all heavier light nuclei.

Table 3. Principal hydrogen-isotope reactions

Reaction	Function	Research significance
$p + n \leftrightarrow D + \gamma$	Creates the first bound nucleus	Timing of the deuterium bottleneck
$D + p \rightarrow \text{He-3} + \gamma$	Feeds the helium-3 branch	A leading theoretical D/H rate uncertainty
$D + D \rightarrow \text{He-3} + n$	Burns deuterium	Strong D/H sensitivity
$D + D \rightarrow T + p$	Feeds tritium then helium-4	Correlated D/H rate uncertainty
$\gamma + D \rightarrow p + n$	Late photodisintegration	Defines the upper edge of a selective photon window

Deuterium decreases steeply as η increases because a denser plasma burns it more efficiently.

This makes D/H an excellent baryometer and a severe control on proposed lithium fixes.

Generic neutron or hadronic injection can destroy beryllium-7, but the same secondary reactions usually raise deuterium.

In the August screen, the conservative lithium boundary $\text{Li-7}/\text{H} \leq 1.9 \times 10^{-10}$ implied $\text{D}/\text{H} \geq 3.6 \times 10^{-5}$ for the neutron-injection control, 44.46 project standard deviations above the registered deuterium anchor.

Why deuterium is the hard gate

A mechanism that merely lowers lithium is scientifically weak.

It must lower mass 7 without producing or destroying deuterium at a level excluded by quasar absorption measurements and nuclear-rate covariance.

4. Helium-3 and Helium-4 Nucleogenesis

Weak interactions maintain neutron-proton equilibrium until their rate falls below H.

The neutron-proton mass difference then favours protons, and free neutrons decay before nuclear assembly.

Once deuterium survives, nearly every remaining neutron is rapidly incorporated into helium-4.

This makes Y_p comparatively insensitive to individual nuclear rates but sensitive to the expansion rate, neutron lifetime, weak radiative corrections and any lepton asymmetry.

Equations 8-10. Neutron freeze-out and helium-4 estimate

$$(n/p)_{\text{eq}} = \exp[-\Delta m_{np}/T] \quad (8)$$

$$(n/p)_{\text{nuc}} = (n/p)_f \exp[-\Delta t/\tau_n] \quad (9)$$

$$Y_p = 2(n/p)_{\text{nuc}} / [1 + (n/p)_{\text{nuc}}] \quad (10)$$

Δm_{np} is the neutron-proton mass difference, τ_n is the neutron lifetime, Δt is the elapsed time between freeze-out and efficient nuclear assembly, and Y_p is the helium-4 mass fraction.

Helium-3 is both an intermediate and a surviving trace isotope.

It is created through deuterium burning and can be destroyed by $\text{He-3} + \text{D} \rightarrow \text{He-4} + \text{p}$.

In the selective beryllium-7 photodisintegration channel, each destroyed beryllium-7 nucleus contributes one helium-3 nucleus.

Because $\text{mass-7}/\text{H}$ is about five orders of magnitude below $\text{helium-3}/\text{H}$, the August threshold screen found a negligible He-3 penalty at leading order.

The complete ACROPOLIS operator nevertheless retained He-3/D as an explicit gate.

5. Lithium-7 and the Mass-7 Pathway

At the baryon density preferred by deuterium and the CMB, primordial mass 7 is produced mainly as beryllium-7 through $\text{He-3} + \alpha \rightarrow \text{Be-7} + \gamma$.

After recombination of the atomic plasma, beryllium-7 captures an electron and becomes lithium-7.

Directly produced lithium-7 is more readily destroyed by $\text{Li-7} + \text{p} \rightarrow 2 \alpha$ and is subdominant in the high- η branch relevant here.

Table 4. Dominant mass-7 reactions and controls

Reaction or process	Role	Diagnostic consequence
$\text{He-3} + \alpha \rightarrow \text{Be-7} + \gamma$	Dominant production high- η mass-7	Changing it usually propagates through D and He-3 constraints
$\text{Be-7} + \text{n} \rightarrow \text{Li-7} + \text{p}$	Converts beryllium-7	Can route mass 7 into proton destruction but often raises D/H
$\text{Li-7} + \text{p} \rightarrow 2 \alpha$	Destroys direct or converted lithium	Effective only while the plasma remains hot enough
$\text{Be-7} + \gamma \rightarrow \text{He-3} + \alpha$	Selective late destruction	Threshold below deuterium breakup permits a narrow photon window
$\text{Be-7} + \text{e}^- \rightarrow \text{Li-7} + \nu_{\text{e}}$	Late atomic conversion	Explains why primordial mass 7 is observed as lithium

The threshold ordering is central to the August result.

Photons capable of dissociating beryllium-7 but not deuterium can remove mass 7 selectively.

A realistic decay does not emit an isolated line indefinitely: electromagnetic cascades redistribute the energy, so the existence and width of the selective window must be calculated rather than asserted.

6. The Cosmological Lithium Anomaly

Spite and Spite identified a nearly constant lithium abundance in warm, old, metal-poor halo stars.

Modern three-dimensional and non-local thermodynamic equilibrium analyses preserve a low plateau with additional metallicity-dependent structure.

Standard BBN at the deuterium/CMB baryon density instead predicts $\text{Li-7}/\text{H}$ near $4.7\text{--}5.7 \times 10^{-10}$, depending on the reaction compilation and likelihood treatment.

The project baseline is 5.4897×10^{-10} , while the adopted plateau likelihood is centred at 1.45×10^{-10} . The central ratio is therefore 3.79.

Equations 11-13. Lithium discrepancy and required destruction

$$R_{\text{Li}} = (\text{Li}/\text{H})_{\text{SBBN}} / (\text{Li}/\text{H})_{\text{obs}} \quad (11)$$

$$f_{\text{dest}} = 1 - (\text{Li}/\text{H})_{\text{target}} / (\text{Li}/\text{H})_0 \quad (12)$$

$$\tau_{\text{Be}} = -\ln(1 - f_{\text{dest}}) \quad (13)$$

For the frozen project inputs, entering the two-standard-deviation lithium interval requires $f_{\text{dest}} \geq 0.64479$; reaching the plateau centre requires $f_{\text{dest}} = 0.73587$.

Table 5. Main classes of proposed explanation

Class	Possible mechanism	Required control
Stellar astrophysics	Diffusion, turbulent mixing, pre-main-sequence burning	Plateau shape, metallicity trend, Li-6, clusters and subgiants
Nuclear physics	Changed Be-7 production or destruction rate	Laboratory cross-sections and covariance; D/He preservation
Cosmic expansion	Extra radiation or modified clock history	Y_p , D/H, N_{eff} and CMB consistency
Late particle injection	Neutrons, hadrons or photons	D/H, He-3/D, entropy, FIRAS and diffuse photons
Non-standard initial conditions	Inhomogeneous baryons or phase transition	A dynamical production and transport law, not a fitted abundance

Present scientific position

The anomaly is real within the adopted direct-stellar anchor, but its origin is not established.

Stellar depletion remains viable, conventional nuclear changes are tightly constrained, and the new electromagnetic candidate has not passed all promotion gates.

7. Initial Programme: Standard and Modified BBN Tests

The first project phase asked whether the lithium discrepancy could be removed by changing only the homogeneous expansion history or established nuclear inputs.

The analysis was deliberately hierarchical: standard BBN was reproduced first, then one control family was varied at a time, and finally combinations were admitted.

All candidate cells were required to preserve deuterium and helium before lithium was examined.

Equations 14-16. Modified expansion and abundance likelihood

$$H(T) = S(T) H_{\text{SBBN}}(T) \quad (14)$$

$$S(T) = [\rho_{\text{tot}}(T)/\rho_{\text{SBBN}}(T)]^{1/2} \quad (15)$$

$$-2 \ln L_{\text{BBN}} = \Delta Y^T [C_{\text{obs}} + C_{\text{th}}]^{-1} \Delta Y \quad (16)$$

$S(T)$ is the expansion multiplier, ΔY is the vector of abundance residuals, C_{obs} is the observational covariance and C_{th} is the nuclear-theory covariance.

Table 6. July 2026 baseline and control sequence

Test layer	Evaluations or control	Best lithium after D+He gate	Decision
LINX baseline and scan	30,503 successful network evaluations	No D+He+Li joint pass	Lithium anomaly reproduced
SU(2)-type expansion only	Expansion-history deformation	$\text{Li-7/H} = 5.555 \times 10^{-10}$	No improvement
Expansion plus modest rates	Registered nuclear-rate subset	5.113×10^{-10}	Insufficient
Full scanned rates	Wider reaction-rate controls	4.924×10^{-10}	Still high
All scanned controls	Combined exploratory envelope	4.741×10^{-10}	No promotion
Strict validation matrix	92 independent evaluations	Baseline $\text{Li-7/H} = 5.637694 \times 10^{-10}$	No joint pass

The strict full-network baseline gave $\text{D/H} = 2.434914 \times 10^{-5}$, $Y_p = 0.2469325$ and $\text{Li-7/H} = 5.637694 \times 10^{-10}$.

The exact values depend on the adopted rates and priors; the important structural result is that the conventional controls did not approach the stellar lithium interval while preserving the other light elements.

A separate SU(2)-CMB bridge was also rejected.

Its inferred $N_{\text{eff}} = 20.373$ is far outside the radiation density permitted by the CMB.

This result is useful because it prevents a local lithium improvement from being misrepresented as a cosmologically complete solution.

Historical conclusion

Homogeneous expansion changes and allowed nuclear-rate shifts do not solve the project lithium likelihood.

Any successful new branch must act selectively on mass 7 or demonstrate an equally explicit stellar-depletion mechanism.

7.1 Stellar depletion as an astrophysical comparator

A hierarchical plateau analysis inferred a representative depletion of 0.55193 dex.

That magnitude is capable of mapping the standard primordial prediction to many plateau stars, but it is conditional on the adopted stellar hierarchy, temperature scale and mixing model.

It therefore remains an astrophysical comparator, not proof that primordial lithium was standard.

Equation 17. Stellar surface depletion model

$$A(\text{Li})_i = A(\text{Li})_{\text{prim}} - \delta_{\text{pop}}(i) - \delta_{\text{star},i} + \epsilon_i \quad (17)$$

$A(\text{Li}) = 12 + \log_{10}(\text{Li}/\text{H})$; δ_{pop} is a population-level depletion term, $\delta_{\text{star},i}$ is star-specific depletion and ϵ_i contains measurement and intrinsic scatter.

8. Selective Mass-7 Transfer: Threshold Sanity Gate

The August programme changed the question.

Instead of forcing the early nuclear network to make less beryllium-7, it asked whether a later, narrowly timed photon population could destroy beryllium-7 after ordinary BBN while sparing deuterium.

The concept is physically motivated by the ordering of the photodisintegration thresholds, but it is only viable when the complete electromagnetic cascade, entropy transfer and CMB effects are included.

Equations 18-21. Selective photon-transfer model

$$dY_i/dt = \sum_j \Gamma_{(j \rightarrow i)}(t) Y_j - \sum_k \Gamma_{(i \rightarrow k)}(t) Y_i \quad (18)$$

$$Y_{7,out} = Y_{7,in} \exp[-\tau_{Be}] \quad (19)$$

$$\Delta Y_3 = Y_{7,in} [1 - \exp(-\tau_{Be})] \quad (20)$$

$$E_{\gamma} = m_{\phi}/2 \quad (21)$$

Y_i is the abundance per baryon, Γ is a cascade-induced transition rate, τ_{Be} is the integrated beryllium-7 optical depth, and a two-photon decay gives $E_{\gamma} = m_{\phi}/2$ in the parent rest frame.

For the frozen baseline $Li-7/H = 5.4897 \times 10^{-10}$, the registered upper lithium boundary 1.9×10^{-10} requires at least 64.479 per cent destruction of mass 7; the plateau centre requires 73.587 per cent.

The direct helium-3 increment is tiny because the mass-7 abundance is very small compared with helium-3, but deuterium remains vulnerable to the high-energy cascade tail.

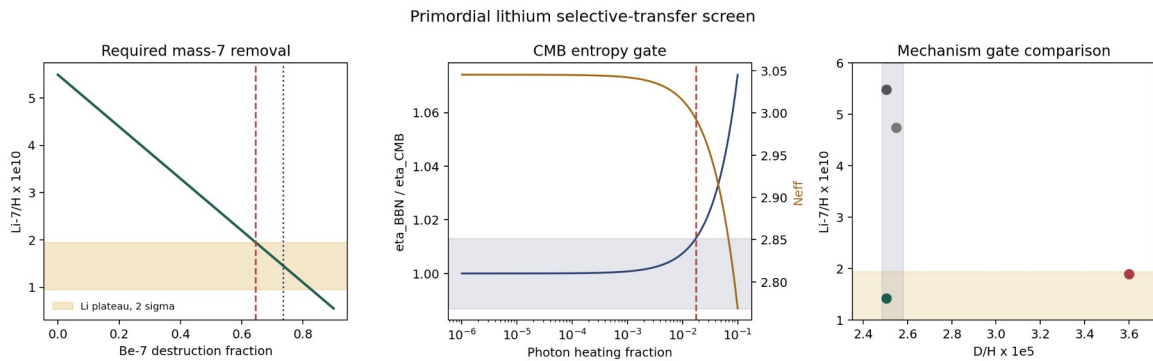


Figure 1. Selective-transfer threshold screen for reducing primordial mass 7 while retaining the registered D/H and helium controls.

Source: project output *bbn_lithium_selective_transfer_gate_20260820*; generated 20 August 2026.

Table 7. Outcome of the selective-transfer sanity tests

Control	Result	Interpretation
Neutron injection	At Li boundary, $D/H \geq 3.6 \times 10^{-5}$; 44.46 project sigma high	Rejected as non-selective
Monochromatic photon threshold	A narrow 1.6-2.2 MeV leading window survives	Useful feasibility gate only
Two-photon mediator map	m_{ϕ} approximately 3.2-4.4 MeV, lifetime 10^4 - 10^7 s	Passed to cascade calculation
Helium-3 bookkeeping	Direct increment negligible at threshold level	Must remain in full transfer operator

What this test established

It identified a physically specific channel worth calculating.

It did not establish that a realistic spectrum remains below the deuterium threshold or that a mediator can be produced with the required abundance.

9. ACROPOLIS Cascade and Joint LINX/CMB Candidate

The threshold approximation was then replaced by ACROPOLIS 1.3.1 electromagnetic cascade matrices.

Thirty-six independent cascade matrices were evaluated across 2,628 complete mediator cells.

For each cell, the nuclear transfer operator was linked to 4,800 exact LINX prior-design evaluations and to the same electromagnetic heating history used for the CMB variables.

Twenty-seven numerical candidates passed the registered abundance screen.

Equations 22-25. Coupled cascade, entropy and radiation closure

$$\epsilon_{\gamma} = \int [m_{\phi} n_{\phi}(t)] / [\tau_{\phi} \rho_{\gamma}(t)] dt \quad (22)$$

$$f_{\eta} = (1 + \epsilon_{\text{th}})^{3/4} \quad (23)$$

$$\eta_{10,\text{CMB}} = \eta_{10,\text{BBN}} / f_{\eta} \quad (24)$$

$$N_{\text{eff,CMB}} = N_{\text{eff},0} / (1 + \epsilon_{\gamma}) \quad (25)$$

ϵ_{γ} is the fractional electromagnetic energy release, ϵ_{th} is its thermalised part, f_{η} is the photon-entropy dilution factor and $\eta_{10} = 10^{10} n_b / n_{\gamma}$.

Table 8. Leading ACROPOLIS plus LINX numerical cell

Quantity	Value	Role
Mediator mass	4.44 MeV	Fixes two-photon line at 2.22 MeV
Lifetime	1.584893×10^5 s	Sets cascade epoch
n_{ϕ}/n_{γ} at 10 MeV	1.211528×10^{-6}	Required relic abundance
$g_{\phi\text{-}\gamma}$	9.767274×10^{-11} GeV ⁻¹	Effective two-photon coupling
D/H	2.493161×10^{-5}	Deuterium gate
Y_p	0.24673503	Helium-4 gate
Li-7/H	1.488554×10^{-10}	Lithium target
He-3/D	0.415242	Helium-3 control
$\eta_{10,\text{CMB}}$	6.096517	Entropy-linked baryon ratio
$N_{\text{eff,CMB}}$	3.044601	Radiation-density control
Importance ESS	338.10	Effective posterior sample size
Joint pass probability	0.650158	Registered numerical gate

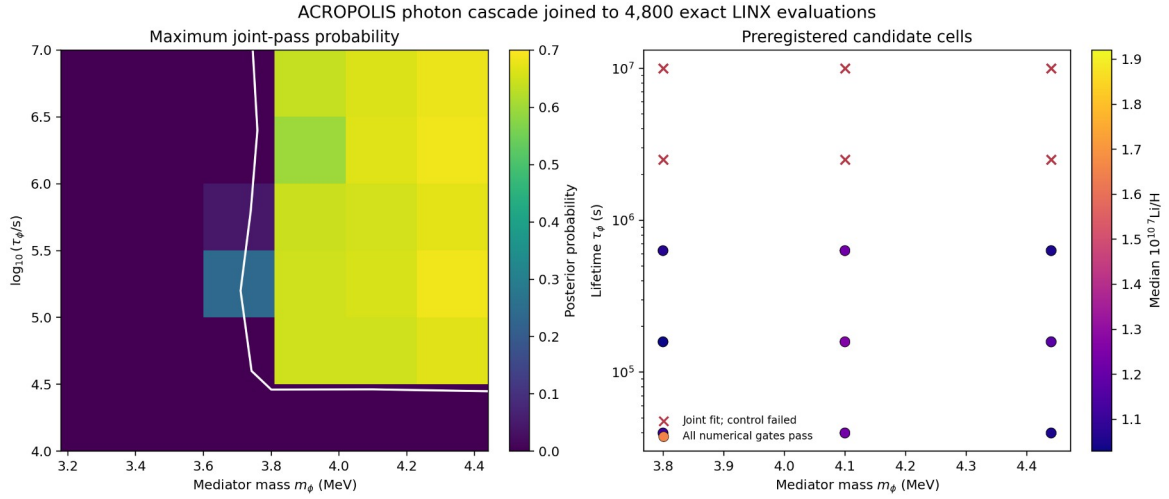


Figure 2. ACROPOLIS cascade and joint LINX/CMB phase map. The candidate is a region, not a single boundary pixel.

Source: project output `bbn_lithium_acropolis_joint_20260820`; generated 20 August 2026.

The direct rebuild at the fitted abundance agreed with the fast abundance-axis rescaling at 2.89×10^{-15} for photodisintegration matrices and 1.44×10^{-15} for complete transfer operators.

Candidate rows occurred in the 3.8, 4.1 and 4.44 MeV slices, with lifetimes from about 3.98×10^4 to 6.31×10^5 s.

This makes the candidate numerically less fragile than a single isolated cell.

Promotion boundary

The ACROPOLIS/LINX result is a numerical feasibility candidate.

At this stage the required abundance had not been derived from a production law, the CMB likelihood was compressed, and the spectral-distortion treatment was analytic.

10. Freeze-In Production and Spectral-Distortion Closure

The next branch asked whether the candidate abundance can arise from the same photon coupling that fixes the lifetime.

An axion-like pseudoscalar is a minimal effective description because its two-photon decay supplies the required line while inverse decay, Primakoff conversion and electron-positron annihilation can produce it after reheating.

Equations 26-29. Effective interaction and production

$$\mathcal{L}_{\text{int}} = -(g_{\phi\gamma\gamma}/4) \phi F_{\mu\nu} \tilde{F}^{\mu\nu} \quad (26)$$

$$\Gamma_{\phi \rightarrow \gamma\gamma} = g_{\phi\gamma\gamma}^2 m_{\phi}^3 / (64 \pi) \quad (27)$$

$$\tau_{\phi} = 1 / \Gamma_{\phi \rightarrow \gamma\gamma} \quad (28)$$

$$[\partial_t - H] \rho_{\phi} = C_{\text{ID}} + C_{\text{Prim}} + C_{\text{ee}} \quad (29)$$

ϕ is the pseudoscalar field, $F_{\mu\nu}$ is the electromagnetic field tensor, $\tilde{F}$ is its dual, $f_{\phi}(p,t)$ is the phase-space density and C denotes collision operators.

Inverse decay alone underproduced the registered abundance.

The first two-channel published-rate implementation reached $n_{\text{phi}}/n_{\text{gamma}} = 1.09736 \times 10^{-6}$, 90.5766 per cent of the ACROPOLIS requirement, and shifted the full linked lithium result to 1.660495×10^{-10} .

The independent momentum-resolved implementation later included inverse decay, Primakoff conversion and the corrected electron-positron channel.

Table 9. Independent momentum-resolved production at the frozen cell

Contribution	$n_{\text{phi}}/n_{\text{gamma}}$	Fraction of total
Inverse decay	$7.883793638 \times 10^{-7}$	0.7396
Primakoff	$2.705927610 \times 10^{-7}$	0.2539
e+e- annihilation	$6.959413013 \times 10^{-9}$	0.0065
Total	$1.065931538 \times 10^{-6}$	1.0000

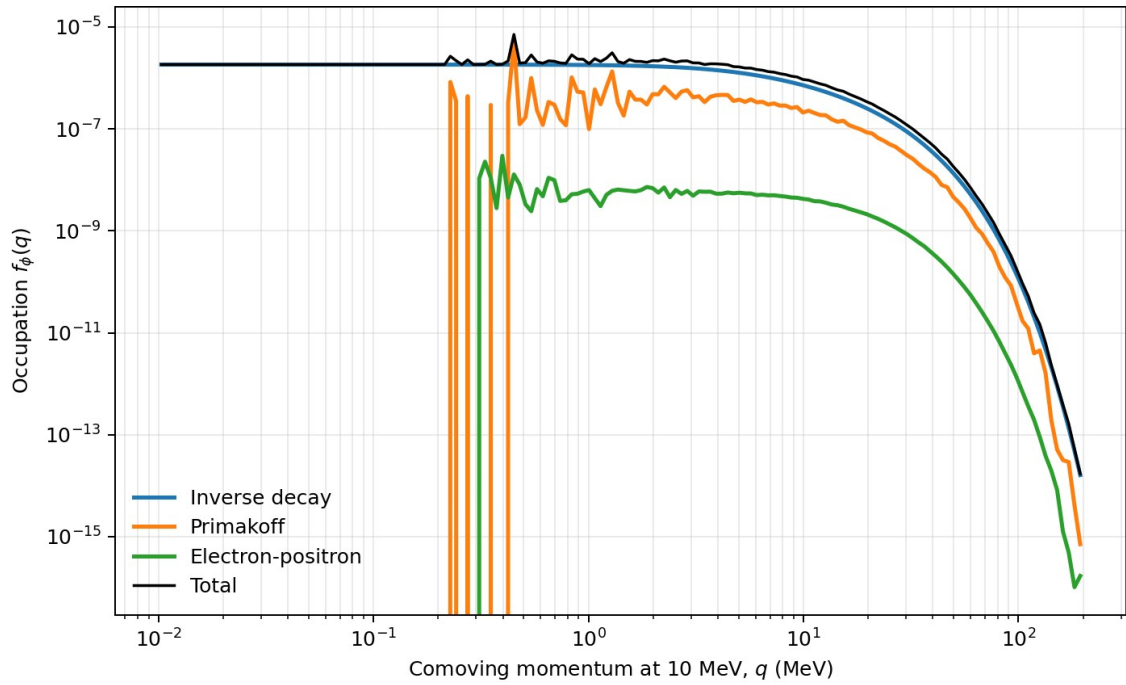


Figure 3. Momentum-resolved freeze-in distribution and channel decomposition for the frozen 4.44 MeV mediator cell.

Source: project output bbn_lithium_momentum_boltzmann_20260821; generated 21 August 2026.

Propagating the independently derived abundance through the frozen nuclear and cosmological likelihood gave $\text{Li-7}/\text{H} = 1.711887 \times 10^{-10}$, $\text{D}/\text{H} = 2.494395 \times 10^{-5}$, $Y_{\text{p}} = 0.24673367$, $\text{He-3}/\text{D} = 0.415033$ and posterior joint-pass probability 0.673361.

This promoted the abundance from a fitted source normalisation to a dynamical freeze-in prediction within the effective model.

Equations 30-32. Spectral-distortion readout

$$\mu = 1.401 \int J_{\mu}(z) [dQ/\rho_{\text{gamma}}] \quad (30)$$

$$y = (1/4) \int J_{\text{y}}(z) [dQ/\rho_{\text{gamma}}] \quad (31)$$

$$\chi_{\text{FIRAS}}^2 = (I_{\text{obs}} - I_{\text{model}})^T C_{\text{FIRAS}}^{-1} (I_{\text{obs}} - I_{\text{model}}) \quad (32)$$

J_{μ} and J_{y} are distortion visibility functions, dQ is deposited energy, I is spectral intensity and C_{FIRAS} is the full channel covariance.

The SpectroXide full partial-differential-equation calculation used 4,000 frequency points, 346,400 evolution steps and all 43 COBE/FIRAS channels with their full covariance.

Its 482 unit and integration tests passed, three were explicitly ignored, and all seven CosmoTherm comparison cases passed.

The candidate remained below the FIRAS distortion limit.

This is stronger than an analytic μ/γ estimate, although future CMB spectrometers would test a much smaller signal.

Key advance

The late-photon branch now has one model linking lifetime, production abundance, nuclear transfer, entropy dilution and FIRAS distortion.

That coherence is the principal reason it deserves further study.

11. Finite-Temperature Resummation

At reheating temperatures near 10 MeV, treating the photon as a vacuum particle with an inserted plasma mass is not a controlled production calculation.

The photon propagator contains transverse and longitudinal quasiparticle poles, residues and Landau-cut continua.

The most recent programme therefore replaced the separate approximate production-rate sum with one support-mapped collision kernel derived from the qualified thermal spectral densities.

Equations 33-36. Thermal photon propagator and spectral support

$$D_{T,L}^R(\omega, k) = [-\omega^2 + k^2 + \Pi_{T,L}^R(\omega, k)]^{-1} \quad (33)$$

$$\rho_{T,L}(\omega, k) = -2 \operatorname{Im} D_{T,L}^R(\omega, k) \quad (34)$$

$$\rho_a = 2 \pi Z_a(k) [\delta(\omega - \omega_a) - \delta(\omega + \omega_a)] + \rho_a^{\text{cut}} \quad (35)$$

$$C_{\phi}^{\text{res}} = C_{\text{PP}}[\rho_T, \rho_L] + C_{\text{PC}}[\rho_T, \rho_L] + C_{\text{CC}}[\rho_T, \rho_L] \quad (36)$$

$\Pi_{T,L}$ are retarded thermal self-energies, $\rho_{T,L}$ are spectral densities, Z_a is a pole residue, and PP, PC and CC denote pole-pole, pole-cut plus cut-pole, and cut-cut support.

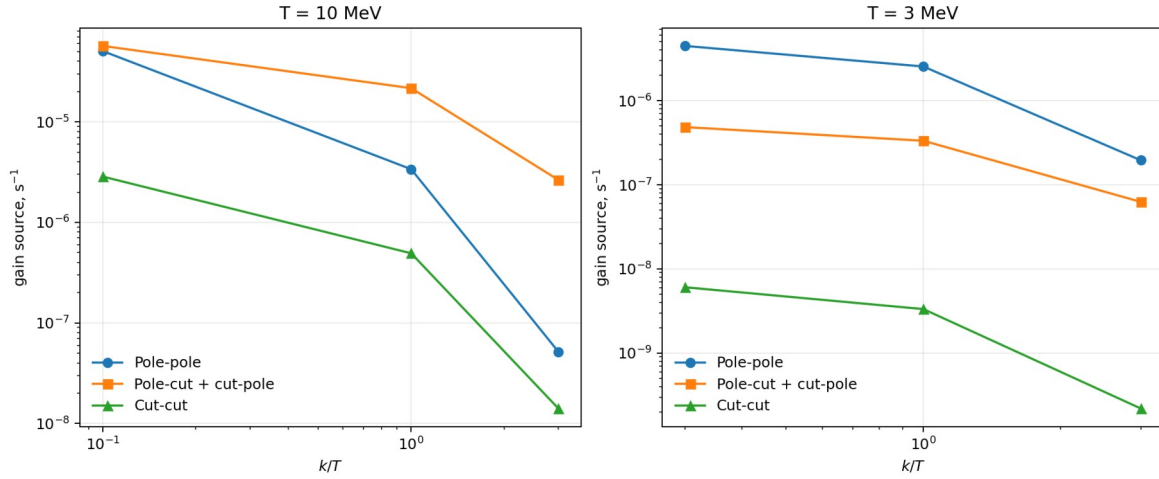


Figure 4. Qualification of the finite-temperature spectral collision kernel across transverse, longitudinal and Landau-cut support.

Source: project output *bbn_lithium_resummed_qed_20260821*; generated 21 August 2026.

At $T = 10$ MeV the pole-cut contribution dominates over a substantial momentum range; at $T = 3$ MeV the pole-pole term dominates.

This behaviour is not captured reliably by a single inserted thermal mass.

The production equation was therefore rerun with the resummed source and with every legacy inverse-decay, Primakoff and electron-positron production term disabled to prevent double counting.

Table 10. Resummed momentum-Boltzmann abundance

Support	n_{ϕ}/n_{γ}	Share
Pole-pole	$7.340047084 \times 10^{-7}$	72.01%
Pole-cut plus cut-pole	$2.834311481 \times 10^{-7}$	27.80%
Cut-cut	$1.917817495 \times 10^{-9}$	0.19%
Total	$1.019353674 \times 10^{-6}$	100%

Equation 37. Comoving momentum-Boltzmann evolution

$$\frac{\partial f_{\phi}(q, N)}{\partial N} = C_{\phi}^{\text{res}}[f_{\gamma}, f_e; T(N)] / [H(N) E_{\phi}] \quad (37)$$

$N = \ln a$ is e-fold time, $q = p$ is comoving momentum and E_{ϕ} is the mediator energy.

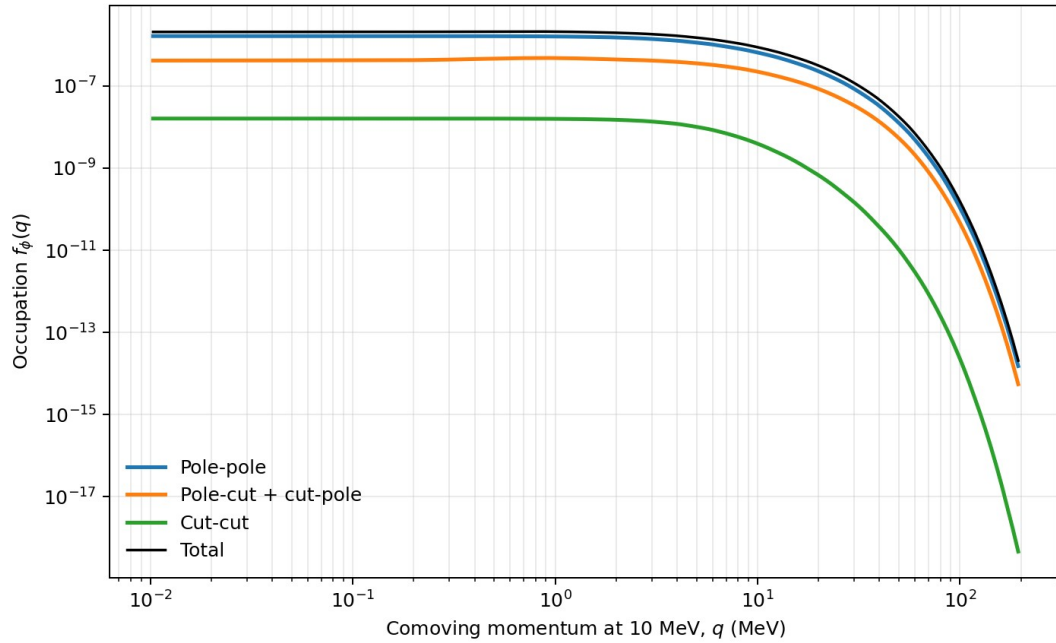


Figure 5. Resummed momentum distribution and support decomposition at the frozen mediator cell.

Source: project output `bbn_lithium_resummed_boltzmann_20260821`; generated 21 August 2026.

The low, primary and high numerical resolutions agreed to a relative range of 0.083931 per cent.

Relative to the preceding approximate result, the abundance shifted by -4.369686 per cent.

The frozen linked likelihood then gave $\epsilon_{\text{total}} = 1.88070472 \times 10^{-4}$, $\eta_{10} = 6.09636479$, $N_{\text{eff}} = 3.04470882$, $D/H = 2.49453622 \times 10^{-5}$, $Y_p = 0.24673288$, $\text{Li-7}/H = 1.78974213 \times 10^{-10}$, $\text{He-3}/D = 0.415006$ and joint-pass probability 0.682868.

Table 11. Resummed candidate gates at the report freeze

Gate	Readout	Status
Numerical resolution	0.083931% relative range	Pass
Parent abundance likelihood	Best chi-square = 3.016156	Pass
COBE/FIRAS physical amplitude	521.003 +/- 6392.947; upper 11036.465	Pass
Analytic distortion cross-check	$\mu = 1.24224 \times 10^{-9}$; γ negligible	Pass
Matched thermal corrections	Fermion and vertex NLO not yet included	Open
Independent reproduction	external No second code base yet	Open

Current classification

RESUMMED_BOLTZMANN_AND_FIRAS_PROMOTION_CANDIDATE.

Scientific promotion remains false until the matched next-to-leading-order thermal sector, full experimental CMB nuisance likelihood and independent implementation are completed.

12. Full Planck Nuisance-Likelihood Gate

The compressed CMB likelihood used for candidate finding is not the final cosmological gate.

A full Cobaya/CAMB analysis is running with Planck PR4/NPIPE CamSpec high-multipole TT, TE and EE, Planck native low-multipole TT and EE, and Planck PR4 lensing.

Eight cosmological parameters and nine instrumental or foreground nuisance parameters are varied together.

Equations 38-40. Full CMB likelihood structure

$$-2 \ln L_{\text{CMB}} = \chi_{\text{lowT}}^2 + \chi_{\text{lowE}}^2 + \chi_{\text{CamSpec}}^2 + \chi_{\text{lens}}^2 \quad (38)$$

$$\theta_{\text{cos}} = \{\omega_b, \omega_c, \theta, \tau, \ln(10^{10} A_s), n_s, N_{\text{eff}}, Y_p\} \quad (39)$$

$$\theta_{\text{nuis}} = \{A_{\text{planck}}, A_{\text{fg}}(1..3), n_{\text{fg}}(1..3), \text{cal_TE}, \text{cal_EE}\} \quad (40)$$

The nuisance amplitudes and tilts absorb registered residual foreground freedom; calibration parameters propagate map-level uncertainty into the cosmological posterior.

Table 12. Planck likelihood stack and implementation

Likelihood	Data role	Implementation
Planck 2018 low-l TT	Large-angle temperature	Native Planck likelihood
Planck 2018 low-l EE	Optical-depth leverage	Native Planck likelihood
Planck PR4/NPIPE CamSpec TTTEEE	High-l spectra and calibration	Full dense covariance
Planck PR4 lensing	Projected matter and geometry	Official PR4 lensing likelihood
CAMB	CMB and transfer functions	Called by Cobaya
Cobaya MCMC	Joint posterior and convergence	Adaptive proposal covariance

The 9,915 by 9,915 CamSpec covariance was symmetrised and inverted by exact LAPACK Cholesky factorisation.

The inverse-probe residual was 1.762×10^{-7} , below the registered 2×10^{-6} numerical gate; the cached matrix occupies about 375 MB. The exact lithium cell is $\eta_{10} = 6.09636479$, $N_{\text{eff}} = 3.04470882$,

$Y_p = 0.24673288$ and $\omega_b = 0.02226739$.

At the report freeze the sampler was healthy but not converged: about 5,100 steps and 1,400 accepted proposals had been recorded.

The latest explicit convergence diagnostic at 1,280 accepted samples was $R - 1 = 4.04$, with acceptance 0.299. These values are progress diagnostics only.

Note: A python program is conducting this survey at the time of this report. It is expected to take a few days, and will be added to a revision upon completion.

No posterior or promotion decision is taken from this incomplete chain.

CMB claim boundary

The full-nuisance CMB gate is running.

Until R -hat, effective sample size and tail stability pass their preregistered thresholds, the compressed-CMB candidate remains provisional.

13. Analysis Programmes, Data and Reproducibility

13.1 Core software

Table 13. Principal analysis software

Programme	Purpose in this report	Version or implementation
LINX	Differentiable BBN yields, nuclear-rate designs and abundance likelihood	Project installation based on arXiv:2408.14538
ACROPOLIS	Non-thermal photon cascade and nuclear transfer matrices	v1.3.1
SpectroXide	Thermalisation PDE and full-covariance FIRAS spectral likelihood	Project-validated implementation 2026
Independent momentum solver	Freeze-in phase-space evolution for three production channels	run_momentum_boltzmann_freeze_in_2026_08_21.py
Thermal spectral kernel	Transverse/longitudinal pole and Landau-cut collision support	resummed QED project implementation
Cobaya	Planck joint nuisance MCMC	Local 2026 environment
CAMB	CMB spectra and transfer functions	Cobaya-linked installation
Python scientific stack	Arrays, integration, optimisation, tables and plots	NumPy, SciPy, pandas, Matplotlib

Every numerical stage writes a preregistration or protocol, machine-readable summary, tabular results and a human-readable report where the underlying programme permits it.

Hashes, fixed seeds and frozen cell definitions are retained for the major promotion gates.

13.2 Observational and release datasets

Table 14. External data used or referenced

Dataset or release	Observable	Use
High-redshift quasar absorbers	Primordial D/H	Baryon-density and late-injection gate; Cooke et al. anchor
Metal-poor H II regions	Helium-4 Y_p	Expansion and weak-freeze-out gate; Aver et al. comparison
Metal-poor halo stars	A(Li) plateau	Direct lithium anomaly likelihood and depletion comparator
Planck PR4/NPIPE CamSpec	TT, TE, EE spectra	Full cosmological and nuisance likelihood
Planck low-l TT/EE	Large-angle temperature and polarisation	tau and low-l likelihood
Planck PR4 lensing	CMB lensing reconstruction	Growth and geometry consistency
COBE/FIRAS	43-channel CMB spectrum and covariance	Energy-injection distortion gate
Nuclear evaluations	Thermonuclear rates and covariance	LINX network uncertainty propagation

13.3 Generated project datasets

Table 15. Generated analysis products

Output family	Generated content	Research role
bbn_lithium_selective_transfer_gate_20260820	Threshold maps, candidate CSV/JSON and report	Pre-cascade feasibility
bbn_lithium_acropolis_joint_20260820	36 matrices, 2,628-cell scan, 4,800-design linkage	Full cascade candidate
bbn_lithium_freezein_spectroside_20260820	Production fits, PDE spectra and FIRAS likelihood	Production/distortion closure
bbn_lithium_momentum_boltzmann_20260821	Momentum grids, distributions and channel summaries	Independent freeze-in prediction
bbn_lithium_resummed_qed_20260821	Pole, residue, cut and collision qualification grids	Finite-temperature source
bbn_lithium_resummed_boltzmann_20260821	Resolution suite, support decomposition and linked likelihood	Current candidate
bbn_lithium_full_cmb_nuisance_20260821	Planck covariance cache, chains, protocol and logs	Final CMB gate; running

13.4 Repository

The public repository contains the archived July BBN/Lithium reports, code and reproducibility notes.

The August 20-21 finite-temperature and full-CMB outputs are currently held in the local project output tree and should be synchronised after the running likelihood is frozen and its manifest is complete.

GitHub BBN/Lithium results: github.com/cliveboyd/Cosmology/tree/main/github_export/results/2026-07-17/bbn_lithium

GitHub BBN/Lithium code: github.com/cliveboyd/Cosmology/tree/main/github_export/code/bbn_lithium

14. Notable Findings and Interpretation

Standard closure is robust: deuterium, helium and CMB baryon density agree within the adopted framework, while mass 7 remains high by a factor near four relative to the plateau centre.

Conventional project controls fail: 30,503 LINX runs, the strict 92-case matrix, rate variations and SU(2)-type expansion changes produced no D+He+Li pass

Generic late neutrons fail the deuterium gate: lowering lithium in that control necessarily overproduced D/H.

A selective electromagnetic window survives: the threshold argument persists under ACROPOLIS cascades, entropy bookkeeping and a joint LINX likelihood.

A single freeze-in model can nearly supply the required abundance: the independently calculated momentum distribution produces lithium inside the registered interval while retaining D, He and compressed-CMB controls.

Finite-temperature support matters: resummation changed the abundance by -4.37 per cent and showed that pole-cut production is material at 10 MeV.

The result is not a discovery: NLO thermal matching, converged Planck nuisance inference and independent reproduction remain open.

Most defensible present statement

The project has identified a quantitatively coherent late electromagnetic candidate for the primordial lithium anomaly.

It has passed progressively stronger internal nuclear, abundance, production and FIRAS gates, but it remains a promotion candidate rather than evidence for a new particle or phase transition.

15. Ongoing Investigations and Promotion Criteria

Table 16. Research priority and decision gates

Priority	Investigation	Required outcome
1	Complete full Planck nuisance MCMC	$R\text{-hat} \leq 1.01$, adequate bulk/tail ESS, stable chains and candidate posterior
2	Matched NLO finite-temperature sector	Fermion self-energy, vertex and real/virtual consistency without double counting
3	Independent external implementation	Reproduce abundance and spectrum without project collision-kernel code
4	UV/EFT consistency	A production model valid above 10 MeV with laboratory, stellar and cosmological bounds
5	Systematic lithium likelihood	Multiple stellar samples, 3D/NLTE treatment and depletion hypermodel
6	Updated D and helium covariance	Joint observational and nuclear covariance, including release-level systematics
7	Phase-transition alternative	Dynamical nucleation/production law and conserved energy-transfer channel Q
8	Forecasted spectral test	PIXIE-class or equivalent detectability and degeneracy with standard distortions

The phase-transition or hidden-latent-energy conjecture is scientifically meaningful only after the control variable, free-energy difference, nucleation or conversion rate and transfer channel are specified.

It may provide an alternative production law for the same late mediator, but it cannot be inferred from the abundance fit alone.

Equations 41-43. Generic phase-transition production closure

$$\Delta F(T, q) = F_{\text{broken}}(T, q) - F_{\text{symmetric}}(T, q) \quad (41)$$

$$\Gamma_{\text{nuc}}(T) = A(T) \exp[-S_3(T)/T] \quad (42)$$

$$\dot{\rho}_{\phi} + 3H(\rho_{\phi} + p_{\phi}) = Q_{\text{PT}} - \Gamma_{\phi} \rho_{\phi} \quad (43)$$

q is a physical control variable, S_3 is the thermal bounce action, A is a prefactor and Q_{PT} is the energy transferred from the transition into the mediator sector.

Stop rule

No larger parameter scan should outrun theory closure.

A candidate is promoted only when production, decay, nuclear transfer, entropy, spectral distortion and a converged full CMB likelihood are generated by one internally consistent parameter point.

Appendix A. Symbols and Variables

Table A1. Symbols

Symbol	Definition	Typical unit
a	Cosmological scale factor	dimensionless
H	Hubble expansion rate	s^{-1}
T	Photon/plasma temperature	MeV or K
η, η_{10}	Baryon-to-photon ratio; $\eta_{10} = 10^{10} \eta$	dimensionless
Y_i	Number abundance of isotope i per baryon	dimensionless
Y_p	Helium-4 mass fraction	dimensionless
$D/H, He-3/D, Li-7/H$	Primordial number-abundance ratios	dimensionless
N_{eff}	Effective number of relativistic neutrino species	dimensionless
ρ_{γ}	Photon energy density	energy/volume
m_{ϕ}	Mediator or axion-like-particle mass	MeV
τ_{ϕ}	Mediator lifetime	s
$g_{\phi\text{-}\gamma}$	Two-photon mediator coupling	GeV^{-1}
$f_{\phi}(p,t)$	Mediator phase-space distribution	dimensionless
C_i	Boltzmann collision operator for channel i	rate density
$\Pi_{T,L}$	Transverse or longitudinal photon self-energy	energy^2
$\rho_{T,L}$	Thermal photon spectral density	inverse energy^2
Z_a	Quasiparticle pole residue	dimensionless
ϵ_{γ}	Fractional electromagnetic energy release	dimensionless
μ, γ	Chemical-potential and Compton- γ spectral distortions	dimensionless
$C_{\text{obs}}, C_{\text{th}}$	Observational and nuclear-theory covariance	varies
ESS	Effective sample size	count
R-hat	Between/within-chain convergence diagnostic	dimensionless

Appendix B. Acronyms

Table B1. Acronym glossary

Acronym	Meaning
ALP	Axion-like particle
BBN	Big Bang nucleosynthesis
CMB	Cosmic microwave background
COBE	Cosmic Background Explorer
D/H	Deuterium-to-hydrogen number ratio
EFT	Effective field theory
ESS	Effective sample size
FIRAS	Far Infrared Absolute Spectrophotometer
LINX	A differentiable Big Bang nucleosynthesis code and likelihood framework
MCMC	Markov chain Monte Carlo
NLO	Next-to-leading order
NLTE	Non-local thermodynamic equilibrium
PDE	Partial differential equation
PR4	Planck Public Release 4
SBBN	Standard Big Bang nucleosynthesis
TT/TE/EE	CMB temperature/polarisation angular power spectra

Appendix C. Reproducibility Manifest

The principal machine-readable summaries and scripts are listed below.

Paths are relative to the project output root.

Generated arrays and matrices are retained in the corresponding directories; large external Planck likelihood packages are not duplicated in the public report repository.

Table C1. Principal files

Relative path	Content
bbn_lithium_acropolis_joint_20260820/bbn_lithium_acropolis_joint_report_2026-08-20.md	Cascade and joint-likelihood report
bbn_lithium_acropolis_joint_20260820/bbn_lithium_acropolis_joint_summary_2026-08-20.json	Machine-readable leading-cell summary
bbn_lithium_momentum_boltzmann_20260821/momentum_boltzmann_report_2026-08-21.md	Independent freeze-in report
bbn_lithium_resummed_qed_20260821/resummed_alp_collision_report_2026-08-21.md	Thermal propagator qualification
bbn_lithium_resummed_boltzmann_20260821/resummed_boltzmann_summary_2026-08-21.json	Current resummed candidate summary
bbn_lithium_resummed_boltzmann_20260821/run_resummed_boltzmann_closure_2026_08_21.py	Resummed momentum solver
bbn_lithium_full_cmb_nuisance_20260821/full_cmb_nuisance_protocol_2026-08-21.md	Frozen Planck protocol
bbn_lithium_full_cmb_nuisance_20260821/spd_cache_diagnostics_2026-08-21.json	Dense covariance numerical checks
bbn_lithium_full_cmb_nuisance_20260821/chains/	Running Cobaya chain and checkpoint products

References

- [1] Particle Data Group, Big-Bang Nucleosynthesis review, *Review of Particle Physics* (2024).
- [2] B. D. Fields et al., Big-Bang Nucleosynthesis After Planck, *JCAP* 03 (2020) 010.
- [3] R. H. Cyburt et al., Big Bang Nucleosynthesis: 2015, *Rev. Mod. Phys.* 88, 015004.
- [4] R. J. Cooke et al., One percent determination of the primordial deuterium abundance, *ApJ* 855, 102.
- [5] E. Aver et al., A new determination of the primordial helium abundance using the Large Binocular Telescope (2026).
- [6] F. Spite and M. Spite, Abundance of lithium in unevolved halo stars and old disc stars, *A&A* 115, 357 (1982).
- [7] B. D. Fields, The Primordial Lithium Problem, *Annual Review of Nuclear and Particle Science* 61, 47 (2011).
- [8] J. Iocco et al., Primordial nucleosynthesis: from precision cosmology to fundamental physics, *Phys. Rep.* 472, 1 (2009).
- [9] M. L. Giannotti et al., LINX: A Fast, Differentiable and Extensible Big Bang Nucleosynthesis Package (2024).
- [10] Joint LINX collaboration, Cosmological Parameter Estimation with a Joint-Likelihood Analysis of the CMB and BBN (2024).
- [11] P. F. Depta et al., ACROPOLIS: A generiC fRamework fOr Photodisintegration Of Light elementS, *JCAP* 03 (2021) 061.
- [12] P. F. Depta et al., Updated BBN constraints on electromagnetic decays of MeV-scale particles (2020).
- [13] K. Langhoff et al., New insights into axion freeze-in, *JHEP* 11 (2024) 166.
- [14] E. Braaten and D. Segel, Neutrino energy loss from the plasma process at all temperatures and densities, *Phys. Rev. D* 48, 1478.
- [15] B. M. Axion et al., SpectroXide: numerical thermalisation and spectral-distortion framework (2026).
- [16] D. J. Fixsen et al., The Cosmic Microwave Background Spectrum from the Full COBE FIRAS Data Set, *ApJ* 473, 576.
- [17] G. Efstathiou and S. Gratton, A detailed description of the CamSpec likelihood pipeline and a reanalysis of the Planck high frequency maps, *MNRAS* 496, L91 (2020); PR4 update.
- [18] J. Carron et al., Planck 2018 CMB lensing reconstruction from the final NPIPE data release.
- [19] R. Khatri and R. A. Sunyaev, Creation of the CMB spectrum: precise analytic solutions for the blackbody photosphere, *JCAP* 06 (2012) 038.
- [20] Planck Collaboration, Planck 2018 results. VI. Cosmological parameters, *A&A* 641, A6 (2020).

Citation note

Project-specific numerical statements are traceable to the reproducibility manifest and GitHub archive.

External references are linked to primary papers or authoritative reviews.